RC Plane Modeling Project

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Modeling the flight of a RC Airplane

Goal is to build a model which dynamically shows the behavior of the aircraft as wattage and oncoming airspeed changes.

To do this we need

- Thrust formula
- Drag formula
- Mass → 1.0kg (Very Convenient)
- Find Velocity of the Exhaust (Ve) given watts and airspeed
    - Component of Thrust

Procedure to calculations

$\frac{1}{2}$*P*A_prop*(Ve^2 - Vo^2) → Thrust

$Ve = Base\_Ve + Vo^{2/3}$

$\frac{1}{2}$*C*A_plane*Vo^2 → Drag

(Thrust - Drag)/mass → acceleration

Acceleration*Time_Step → Velocity

Time_Step*Velocity → Displacement

$Time\_Step = 0.01$

Key:

P = Air density = 1.255
A_plane = 0.10
A_prop = 0.025
C = drag coef = 0.10
Ve = Velocity thrust
Vo = Velocity plane

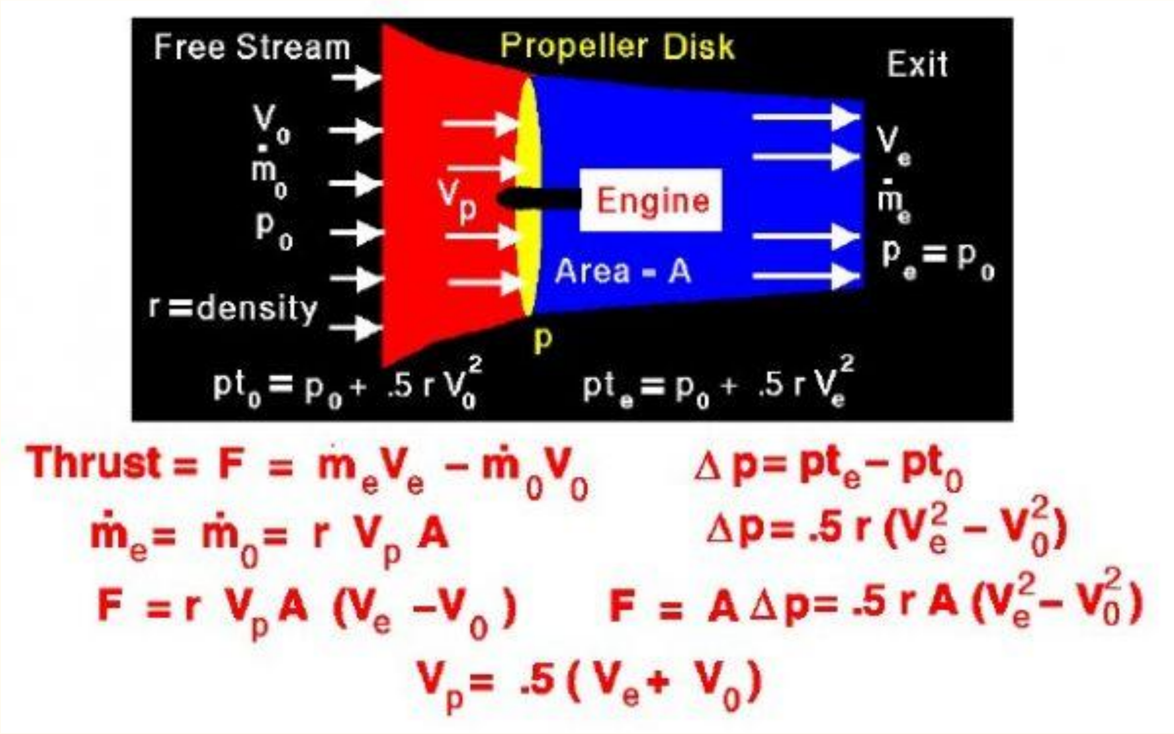
Thrust = 0.5 * p * A * (Ve**2 - Vo**2)

[Propeller Thrust | Glenn Research Center | NASA]

Ve = **BaseVE** + Vo^(⅔)

BaseVe → as Function states

- P = 1.255km/m^3
  - Air Density
- A = 0.025m^2
  - Prop Area
- Vo = **VAR**
  - Speed of Plane

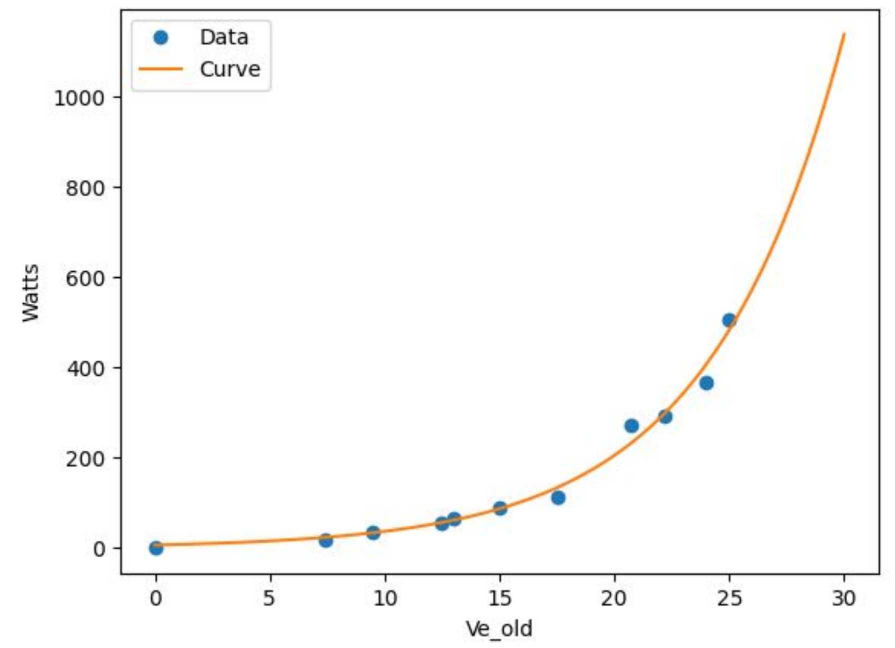


Watts Relation to Ve

5.83805 ln(0.14987 x Watts)
Derived Using SciPy

- Finding the Line of Best Fit
- Take the Inverse

SciPy

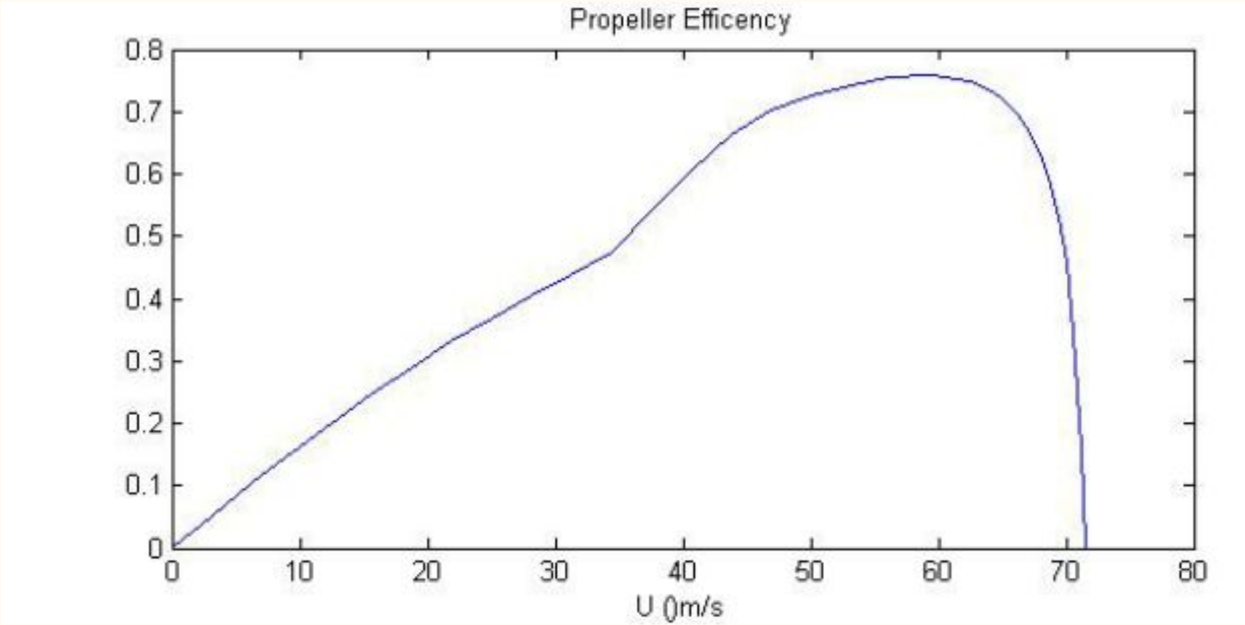


Efficiency Loss

- Issues with Prop. Driven Aircraft

- Propellers lose "efficiency" as the plane move faster

- This is why we use Jets



Wind Tunnel?

Distinctly No Wind Tunnel →

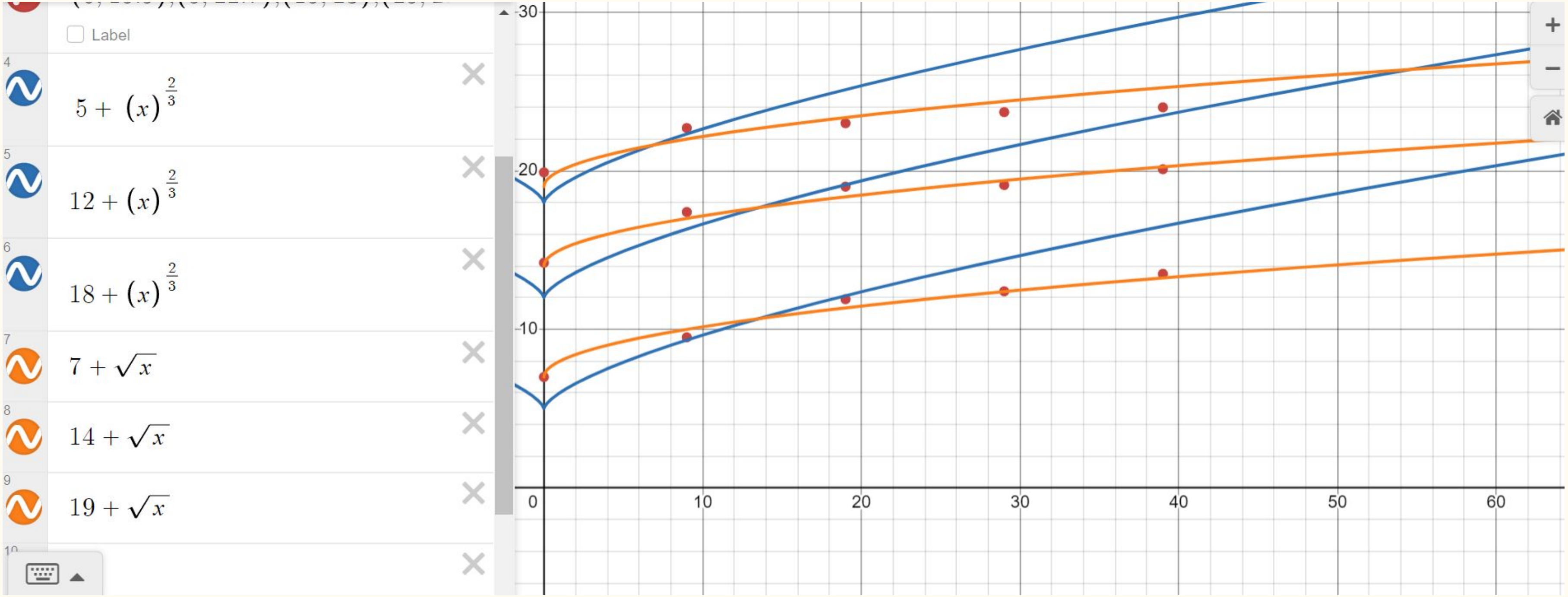
Cabrillo

Gathering Airspeed Data → Anemometer

Plane Car

It's a plane, mounted to a car

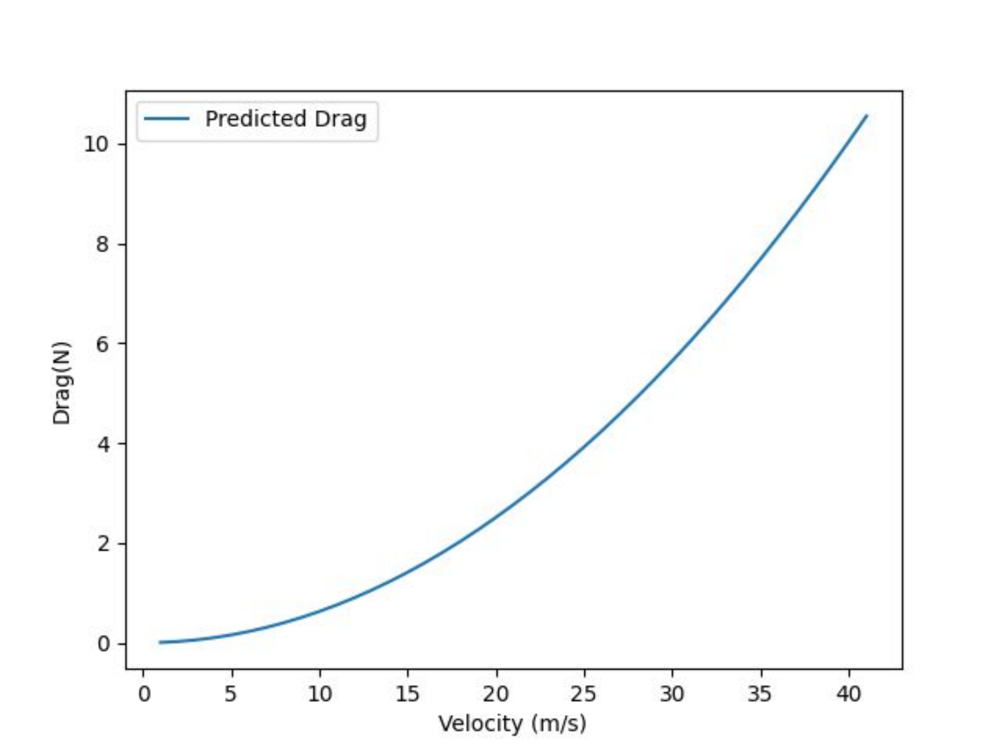
Calculating Velocity Exhaust(Ve) relation to Airspeed(Vo)



Drag

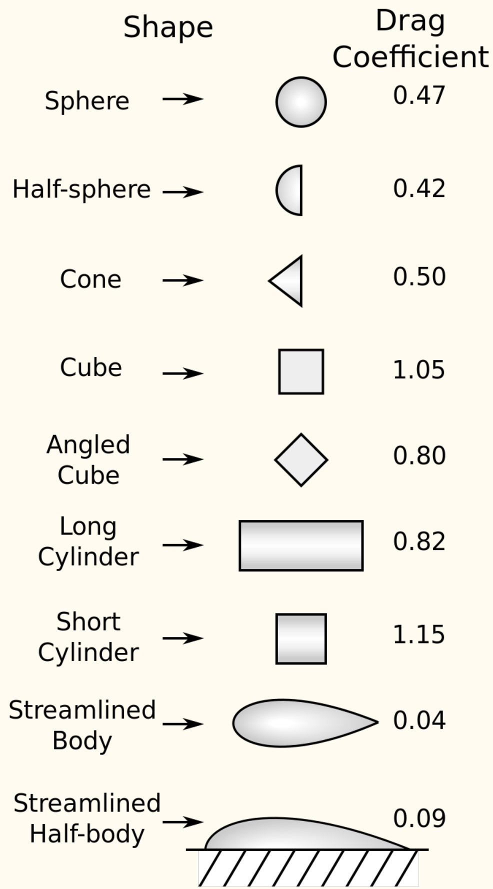
Drag = .5*CpA*Vo^2

- C = 0.1
  - Average Drag Coefficient
  - Pretty Much just an educated guess
- P = 1.255km/m^3
  - Air Density at sea level
- A = 0.1m^2^2
  - Frontal Area of the Plane (pi r)
    - Measured with *Geometry*



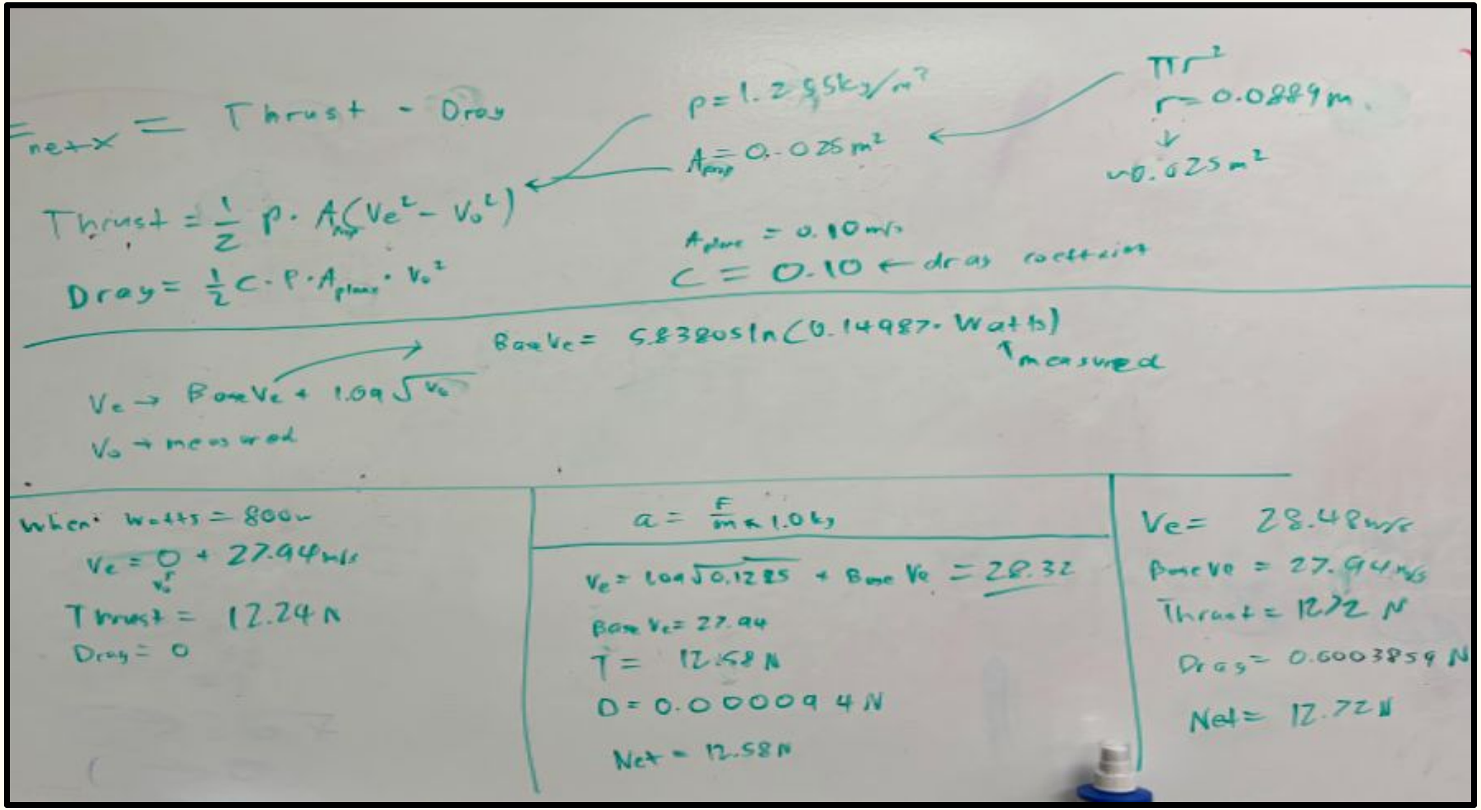
Drag Coefficient Approximation

- 0.04 → Airfoil
- Fuselage
  - Less Efficient
- Thus always higher than 0.04
- 0.10 is quite good, not implausible.
- Model was tested with other coefficients
  - 0.07 - 0.20



Measured Drag Coefficients

Sample Calculation



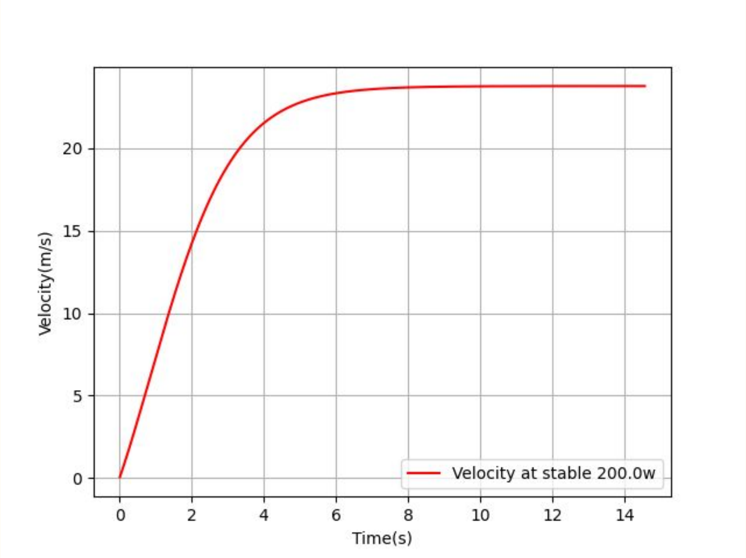
Sample Calculation

Time	X (m)	V (m/s)	A (m/s^2)	F_Net (N)
0.00	0.00	0.00	12.25	12.24
0.01	0.00	0.1225	12.58	12.58
0.02	0.0012	0.248	12.72	12.72

Experiential vs Theory → 200 Watts

Predict. Max: 86.4km/h | 24.3m/s

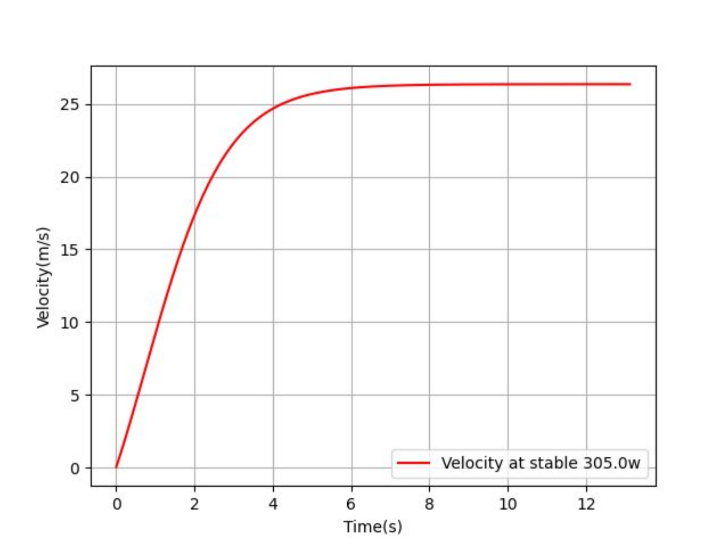
Actual Max: 83.0 km/h | 23.1m/s



Experiential vs Theory → 305 Watts

Predict. Max: 97.2km/h | 27.2m/s

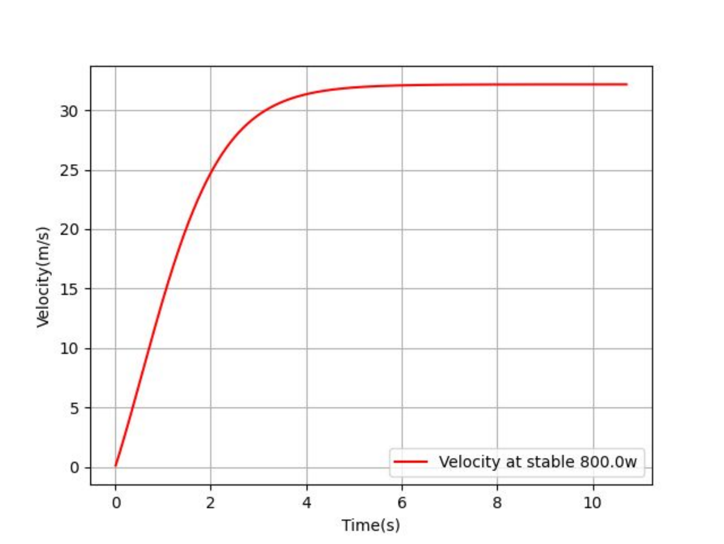
Actual Max: 95 km/h | 25.6m/s



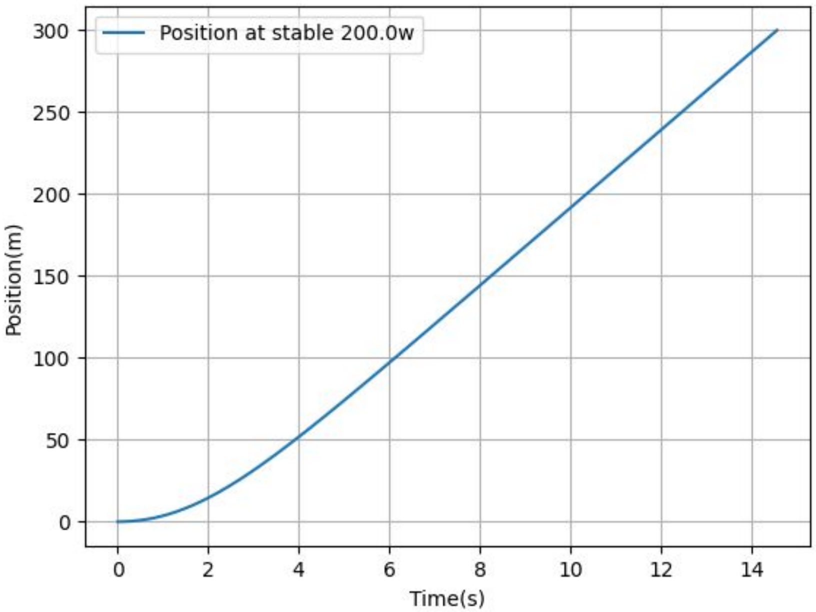
Experiential vs Theory → 800 Watts

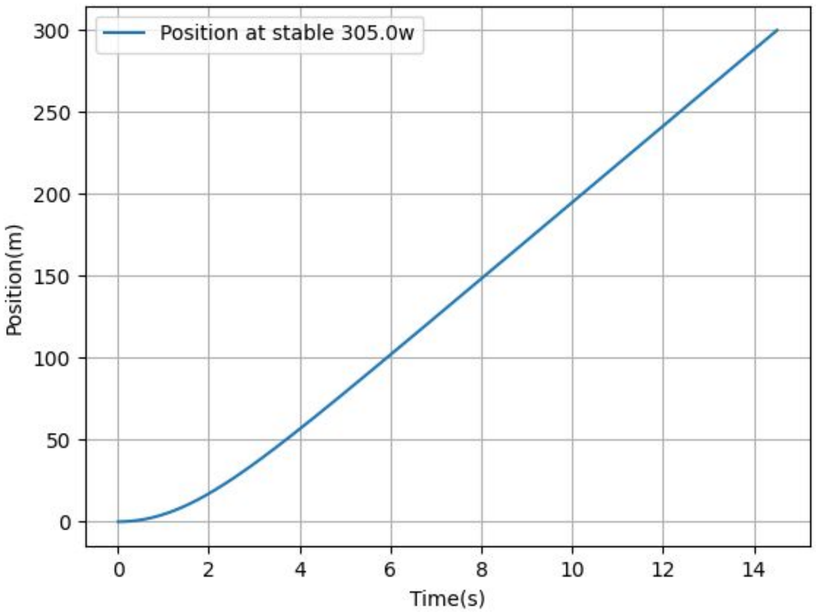
Predict. Max: 118.8km/h | 33.1m/s

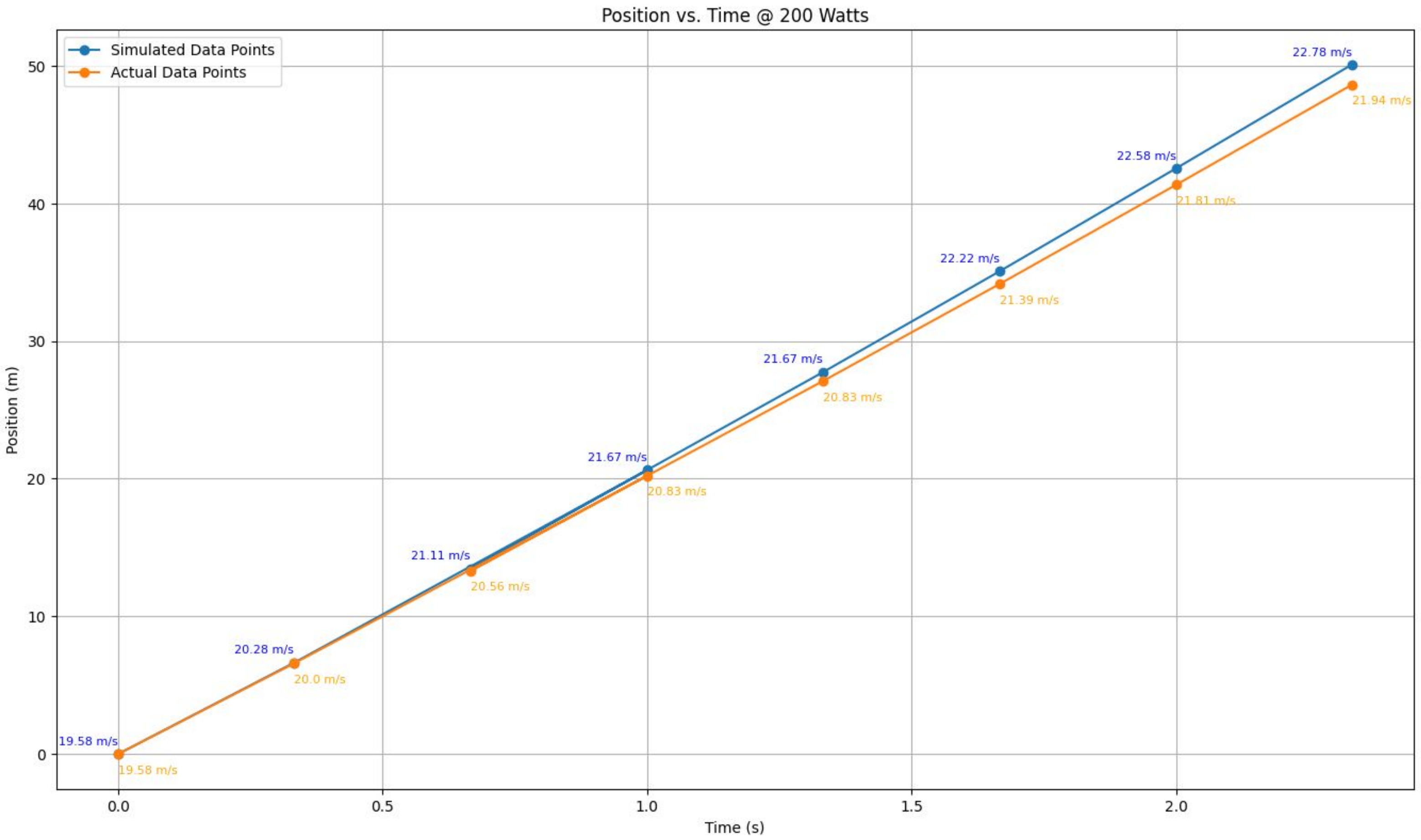
Actual Max: 120km/h | 33.3m/s



Position vs Time







Position vs. Time @ 200 Watts
Simulated Data Points
Actual Data Points
Position (m)
Time (s)
0
10
20
30
40
50
0.0
0.5
1.0
1.5
2.0
19.58 m/s
19.58 m/s
20.28 m/s
20.0 m/s
21.11 m/s
20.56 m/s
21.67 m/s
20.83 m/s
21.67 m/s
20.83 m/s
22.22 m/s
21.39 m/s
22.58 m/s
21.81 m/s
22.78 m/s
21.94 m/s

Demo and Python Script